

Gemini CLI & GitHub Integration Guide

WINDOWS OS ENVIRONMENT DOCUMENTATION

1. Prerequisites

Before initiating the installation of Gemini CLI on your Windows machine, ensure the following native dependencies are configured and properly appended to your System Environment Variables (**PATH**):

- **Node.js**: Version 18.x or higher (LTS version recommended).
- **NPM**: Bundled automatically with your Node.js Windows Installer (MSI).
- **Git for Windows**: Ensure Git Bash or standard Git tools are operational and map correctly within the command environment.

2. Installing Gemini CLI

Follow this exact execution sequence inside a Windows-compatible shell terminal framework.

Environment Elevation

To avoid security blocks or local access level constraints, launch your terminal framework under administrator permissions:

- Press the **Windows Key**, search for **PowerShell** or **Command Prompt**, right-click the application, and select **Run as Administrator**.

Global Installation Execution

Run the global installation command via NPM to fetch the latest binary executable distribution:

```
npm install -g @google/gemini-cli
```

Verifying Installation via Terminal

Validate that the environment variable mapping registers the new command globally across your system path loop:

```
gemini --version
```

3. Connecting GitHub to Gemini CLI

To link your local Windows development system context securely to your remote code repositories, configure a secure authentication bridge using a GitHub Personal Access Token (PAT).

Step 1: Generate a GitHub Personal Access Token

1. Access your standard developer dashboard on **GitHub.com**.
2. Click your user profile portrait avatar located at the top-right corner → **Settings**.
3. On the left structural navigation pane, scroll down to the bottom and click on **Developer settings**.
4. Select **Personal access tokens** → **Tokens (classic)**.
5. Click the **Generate new token** drop-down element, then pick **Generate new token (classic)**.
6. Assign a transparent identity descriptor label (e.g., `gemini-cli-windows-token`).
7. Tick the specific structural check-boxes to apply authorization scopes:
 - `repo` (Grants total read/write privileges over your private and public codebases)
 - `read:user` (Allows structural access to basic profile properties)
8. Click the **Generate token** action button at the base of the portal page.

Critical Security Notice: Immediately copy the alphanumeric token string to your clipboard. Once you navigate away from this screen, GitHub hides this key permanently.

Step 2: Save the Authentication Credentials Locally

Return to your open administrative PowerShell or Command Prompt console window and invoke the handshake login parameters:

```
gemini login --provider github
```

When the prompt asks for your security credentials, right-click into the terminal window (or press `Ctrl + V`) to securely input your copied Personal Access Token string, then hit **Enter**.

Step 3: Confirm Access Privileges

Run a status check command to guarantee that the token profile successfully authenticates past the remote access gateways:

```
gemini auth status
```

Upon a valid connection routine, the command window outputs your verified GitHub profile username alongside an active authorization message.

4. Windows Troubleshooting Guide

- **'gemini' is not recognized as an internal or external command** : This error surfaces when the global NPM installation folder path isn't mapped inside your Windows System Environment Variables. To

resolve this, locate your NPM path (typically `%AppData%\npm`) and manually append it to the system `PATH` via the Windows Advanced System Settings panel.

- **Execution Policy Restriction Errors** : If Windows PowerShell blocks script execution, clear the path permissions by running:
`Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser`
- **Bad Credentials (401 Error)** : The entered authentication token string has expired or was clipped during copy. Re-run `gemini login --provider github` to supply a valid credential string.